The Tritone, the Minor Sixth and the Dissonance Curve: A Reanalysis of Dense Two-Note Ratings

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Abstract

Music teaching treats the tritone, two notes six piano keys apart, as the model of a harsh-sounding (dissonant) pair. The standard acoustic explanation is a curve built on roughness, the harshness arising when overtones lie close together. In my computation that curve predicts the tritone to sound about as pleasant as the minor sixth, eight keys apart, yet listeners prefer the minor sixth. This study measures that disagreement and asks why. I reanalysed published ratings of two-note sounds whose pitch distance was drawn from a continuous range (Marjieh et al., 2024), fitting published curves to the ratings at every distance except those two. For ordinary musical tones, US listeners rated the minor sixth above the tritone by +0.26 times the spread (standard deviation) of a listener's ratings, and roughness predicted none of it. Adding harmonicity, a published measure of how closely the combined overtones resemble a single note, with one setting that I selected, explained a small part. Most of the difference remained (+0.18 in the same unit), detectable at the minor sixth, which listeners liked more than the curves predict, not at the tritone. The data cannot say whether the remainder reflects an incomplete model or learned familiarity.

Keywords: consonance; dissonance curve; tritone; roughness; harmonicity; reanalysis; equal temperament; noise ceiling; cross-cultural; timbre

1 Introduction

The question. Some pairs of musical notes sound pleasant together and others sound harsh. Musicians call the first kind *consonant* and the second *dissonant*. The distance in pitch between two notes is called an *interval*, and it is counted in *semitones*, the step between adjacent keys of a piano. One interval has a special reputation: the *tritone*, two notes six piano keys apart, which music teaching treats as the model of a dissonance. The standard acoustic explanation of dissonance is a *dissonance curve*: a prediction, computed from the physics of the sound, of how harsh each interval should be (Section 2 explains how). That curve does not single the tritone out. In my computation for the sounds analysed in this paper, it predicts that the tritone should sound about as pleasant as the *minor sixth*, the pair eight keys apart, yet listeners prefer the minor sixth. This paper measures that disagreement in published listening data and asks which property of the sound, if any, accounts for it.

Why I ask. The question comes from my own playing. I learned music from a young age and accompanied church services, and for years I used chord progressions (sequences of chords) and improvised by feel, without knowing the theory behind them. A video about dissonance made me want to understand it properly.

What this paper does. The paper builds on the work of others at every step. The listening data were collected by three research groups, and the models of dissonance were proposed by others; Section 2 describes that work, one line of research at a time, and says what it left open. My part is a reanalysis. I put one narrow question to the published ratings, I measure the answer in a way that does not depend on the arbitrary units of the models, and I test the published curves on experiments that were not used to fit them (Section 3). No new listener data were collected.

What it finds, in brief. For ordinary musical tones, listeners in the United States (US) rate the minor sixth clearly above the tritone, and the standard curve predicts none of that difference. A second property of the sound, its *harmonicity* (how closely the combined sound resembles a single note), explains a small part of the difference, by lowering the prediction for the tritone. Most of the difference remains, and it lies at the minor sixth: listeners rate it, like the other consonant intervals, above the curve, while the tritone is not reliably below the curve. Several further ideas that I tested did not help, and the paper reports them as failures. The data cannot decide why the consonant intervals are liked more than the curves predict.

How to read this paper. Sections 2 and 3 give the background and the plan in plain terms, without results. The Data, Models and Method sections give the detail needed to check the work. The Results section reports the main analyses, and the section after it the analyses chosen only after results had been seen (exploratory analyses); the Discussion, Limitations and Conclusion follow. A reader who wants only the argument can read up to the end of Section 3 and then the Conclusion.

2 Research landscape

This section summarises the earlier work that this study rests on: who did it, what they concluded, and what I take from it. I give conclusions and leave numerical detail to the cited papers. Table 1, at the end of this section, lists the studies in order of time. The survey is not systematic: it covers the work I found and used, not everything that has been published on consonance.

Terms needed to read on. Intervals are measured in semitones and in *cents*, hundredths of a semitone. On a piano tuned in the standard way every semitone is the same frequency ratio, and twelve of them make an octave, a doubling of frequency; a pair of notes whose frequencies are in the ratio q is therefore $1200 \log_2 q$ cents apart. Two notes sounded together are a *dyad*; three or more are a chord (where a published data set counts dyads among its chords, I keep its wording). From one to twelve semitones the intervals are called minor second, major second, minor third, major third, fourth, tritone (six semitones), fifth, minor sixth (eight semitones), major sixth, minor seventh, major seventh and octave; the fourth and the fifth thus lie on either side of the tritone. A musical *scale* is the set of pitches that a musical tradition uses; in Western music neighbouring pitches of the scale are a whole number of semitones apart. The studies reanalysed here asked listeners either how pleasant or how consonant a sound was, and I treat the two as the same quantity. A *pure tone* is a single sine wave. A musical note is normally a *complex tone*: a sum of pure tones called *partials* or overtones. In a *harmonic complex tone* the partials lie at whole-number multiples of the lowest frequency, as in the voice and in most pitched instruments; these partials are called harmonics. The set of partials and their strengths is the tone's *spectrum*, and is what makes a flute and a guitar sound different when playing the same note (their *timbre*).

Roughness and the dissonance curve (from 1965). Two pure tones that are close in frequency interfere, and the loudness of the sum fluctuates. This fluctuation is called *beating*; when it is rapid the sound is heard as rough. Plomp and Levelt (1965) and Kameoka and Kuriyagawa (1969) measured how unpleasant two pure tones are as a function of their separation, and Plomp and Levelt related the result to the *critical bandwidth*, the frequency range within which the ear does not fully separate two tones. Hutchinson and Knopoff (1978) turned this into a model for complex tones: take every pair of partials in the sound, compute the *roughness* of that pair, and add the results. Sethares (1993) gave a similar model and showed that its predictions depend on the spectrum of the tones. Plotting the summed roughness of a dyad against its interval gives the dissonance curve of the Introduction: the horizontal axis is the interval, and the vertical axis is the predicted roughness, so that a low point is an interval predicted to sound smooth. For harmonic complex tones such curves have low points at the octave, fifth and fourth and high points near the semitone and the major seventh (Sethares, 2005), which matches much of the familiar ranking of intervals. They are the basis of the roughness account of consonance.

Harmonicity (from 1974). A second line of work holds that a sound is pleasant when its partials, taken together, resemble the harmonics of one note. Terhardt (1974) described how the ear assigns a single pitch, the *virtual pitch*, to a set of harmonics, and Parncutt (1989) built a model of harmony on that idea; Bowling and Purves (2015) argue for the same property on biological grounds. This property is the harmonicity of the Introduction. Harrison and Pearce (2018) and Harrison and Pearce (2020) reviewed the available models and described the consonance of chords by three components: interference between partials (roughness), harmonicity, and *familiarity*, meaning how often a listener has met the chord in music.

Ratings of chords (2012 and 2018). Models need listening data to be tested against. Ratings of chords were collected by Johnson-Laird et al. (2012) and by Bowling et al. (2018), and Eerola and Lahdelma (2021) compared acoustic and cultural predictors across several such data sets. Chord data, however, contain only the intervals of the musical scale, so, as Marjieh et al. (2024) point out, they cannot show what lies between those intervals. By my reading, measures that score an interval by the simplicity of its exact frequency ratio (how small the whole numbers in the ratio are, as in 3/2), of which De Roure (2026) is a recent example, likewise say nothing about the intervals in between.

The tritone in this literature. On a roughness curve for harmonic tones the tritone sits at about the same height as the minor sixth. This statement rests on my own computation: Figure 1, in the Method section, shows it for the Hutchinson–Knopoff model on the sounds analysed in this paper (red dashed line), and Section 7.2 tests it for each spectrum. The literature has a general reply to mismatches of this kind: the curve is an oversimplification, and factors it does not capture decide the rest (Parncutt & Hair, 2011; Harrison & Pearce, 2020; Lahdelma et al., 2022). Parncutt (1989) and Parncutt and Hair (2011) attribute the tritone's dissonance to its being neither smooth nor harmonic, and Parncutt et al. (2019) note that roughness alone cannot order the fourth and the minor sixth.

Dense dyad ratings (2024). Marjieh et al. (2024) removed the restriction to the scale. They asked listeners to rate dyads whose interval was drawn at random from a continuous range of 15 semitones, not from the scale, and repeated the experiment with many tone spectra and with listeners in the United States and in South Korea. I call these data *dense dyad ratings*, because the interval axis is sampled densely. Marjieh et al. concluded that the intervals listeners prefer

Table 1: *Prior work.* The studies this paper builds on, in order of first publication, with the conclusion I take from each and the use made of it here. Terms are explained in the text of this section.

Study	What it concluded or provided	Use in this paper
Plomp and Levelt (1965); Kameoka and Kuriyagawa (1969)	Two pure tones sound least pleasant when they are a small distance apart, a distance tied to the resolution of the ear	basis of every roughness model
Terhardt (1974); Parncutt (1989)	The ear assigns one pitch to a set of overtones; a model of harmony built on that idea	idea behind harmonicity; one of the further properties tested
Hutchinson and Knopoff (1978)	A roughness model for musical notes: add up the roughness of every pair of overtones	the roughness curve tested here
Sethares (1993, 2005)	A similar model; the curve depends on the overtones of the instrument	second roughness curve
Johnson-Laird et al. (2012)	Listeners' ratings of chords of three and four notes	chord analysis
McDermott et al. (2016); Lahdelma et al. (2022)	Preferences for consonance vary with culture and familiarity	the alternative reading of what remains
Bowling et al. (2018)	Listeners' ratings of all chords of two to four notes within an octave	chord analysis; twelve dyads
Harrison and Pearce (2018, 2020)	A review of consonance models; consonance described by roughness, harmonicity and familiarity; a harmonicity model	the harmonicity model used here
Masina et al. (2022)	Roughness combined with a second term for dyads	idea behind one of the further properties tested
Marjieh et al. (2024)	Ratings of dyads at intervals drawn from a continuous range, for many kinds of tone and two countries; a combined model	main data; their model as a comparison

can be reshaped by changing the spectrum of the tones, and they fitted a combination of a revised roughness model (one with an added term for slow beating, which they report listeners to like) and a harmonicity model to their data. They did not examine the tritone and the minor sixth separately, which is the subject of this paper. Separately, and before those data existed, Masina et al. (2022) had combined roughness with a second term for dyads, based on how regularly the combined sound wave repeats.

The open debate. Two readings of consonance are current. In accounts based on hearing alone (psychoacoustic accounts; Harrison and Pearce, 2020; Marjieh et al., 2024) harmonicity is something the ear computes. In the other view harmonicity partly stands in for cultural familiarity, that is, for what listeners have learned from the music around them (Lahdelma et al., 2022; McDermott et al., 2016; Milne et al., 2023). This paper does not settle that debate.

3 This study: the open question and the approach

What remained open. The reply that "other factors decide the rest" is plausible but had not been made quantitative for the tritone. As far as I could find, it had not been reported how large the discrepancy is, whether a better curve removes it, or whether what remains belongs to the tritone at all. These are the three questions of this paper.

Why these data and these steps. Each step of the analysis answers a practical difficulty. *Dense ratings.* Because the interval is sampled continuously, the rating at the tritone can be compared with a curve fitted to all other intervals, so the discrepancy can be estimated without relying on the units of the model, which are arbitrary. *The minor sixth as the reference.* The

roughness curve gives the tritone and the minor sixth nearly the same value, so a difference between their ratings is a difference that roughness does not produce. *Experiments kept out of the fit.* A curve adjusted to an experiment can follow the noise of that experiment, so the curves are judged on experiments that played no part in adjusting them. *Tones with shifted partials.* Because the spectrum varies between experiments, one can ask whether the discrepancy follows the partials (moves to another interval when the partials are moved), as an acoustic mechanism would, or stays with the interval, as a category learned from music ("six semitones") would. *Chords.* The two chord data sets show whether the same pattern appears when the tritone occurs inside a chord. *A limit on what any curve can do.* Ratings are noisy, so the curves are also compared with the noise ceiling, the best fit that the reliability of the ratings allows.

What is mine and what is not. This paper is a reanalysis of public data. No new listener data were collected and I ran no listening experiment. The ratings come from three published studies: the dense dyad ratings of Marjieh et al. (2024) and the chord ratings of Bowling et al. (2018) and Johnson-Laird et al. (2012). The models also come from prior work: the roughness models of Hutchinson and Knopoff (1978) and Sethares (1993), the harmonicity model of Harrison and Pearce (2018, 2020), and the revised roughness model and its combination with harmonicity by Marjieh et al. (2024). The harmonicity model has a setting for how far a partial may be mistuned and still count as harmonic (the Models section defines it). The idea of widening it is not mine either: the code released with Marjieh et al.'s paper defines such a variant but does not use it (Harrison, 2023); the Models section says which values are theirs and which are mine. My contribution is the analysis: (a) a quantity specific to the tritone, namely the difference in rating between the minor sixth and the tritone and the part of it that a curve fails to predict, measured against a curve fitted to all other intervals; (b) an evaluation of the curves on experiments that were not used to fit them, together with a comparison of further candidate properties of the sound, listed before that comparison was run; (c) an exploratory test with stretched spectra (tones whose partials are spaced more widely than harmonics; the tones are those of Marjieh et al.), which asks whether the tritone's low rating moves when the partials move; (d) a comparison of the curves with the noise ceiling; and (e) a check that chord models are scored at the tuning (the exact frequencies of the notes) in which the chords were presented. The chords of Bowling et al. were played in a tuning slightly different from standard keyboard tuning (the Data section describes both), and scoring them at the pitches that were played changes how much of the listeners' ratings the models reproduce (Section 7.7). Each table and figure caption begins with "Prior data" or "Prior work" when it reproduces published material and with "This study" when it shows my analysis of the published ratings; the paragraphs of the Method section are marked in the same way.

4 Data

All data are published ratings; Table 2 summarises them. I describe them in the order in which they were collected.

Chord ratings of Johnson-Laird et al. (2012). Johnson-Laird et al. (2012), of Princeton University and Dartmouth College, asked listeners to rate chords on a seven-point scale from 1 (highly pleasant) to 7 (highly unpleasant), so that, unlike the Bowling ratings described next, a higher value means a less pleasant chord. The listeners came from the Princeton University community and included musicians and non-musicians: 27 rated the three-note chords (their

Table 2: *Prior data.* The three published rating sets reanalysed in this paper. Counts for the dense dyads are those analysed here, from the released files (see text). Equal temperament: the keyboard tuning with twelve equal semitones per octave; just intonation: intervals tuned to exact whole-number frequency ratios. The two chord sets are rated in opposite directions; the chord regressions of Section 7.7 put both on one direction, in which a higher value is more dissonant.

	Johnson-Laird et al. (2012)	Bowling et al. (2018)	Marjieh et al. (2024)
Stimuli	103 chords of three and four notes	298 chords of two to four notes	dyads at random intervals from 0 to 15 semitones
Tuning	equal temperament (my inference; not stated in the article)	just intonation	any interval, not tied to a scale
Listeners	27 (three-note chords) and 39 (four-note chords)	30	1,627 (US) and 122 (South Korea) participant identifiers analysed (counted once for each experiment, so the same person may be counted more than once; see text)
Rating	1 (highly pleasant) to 7 (highly unpleasant)	consonance, four points; in the data a higher value is more consonant (the article describes the scale the other way round; see text)	pleasantness, seven points
Use in this paper	chords (Section 7.7)	chords and twelve dyads (Sections 7.1 and 7.7)	all dyad analyses

Experiment 1) and a different 39 rated the four-note chords (Experiment 2). The published set contains mean ratings for 103 chords, 55 of three notes (triads) and 48 of four notes (tetrachords), arranged so that, in the article's words, the chords "spread over about an octave and a half". The chords were written in the notation program Sibelius and synthesised as piano chords. The article does not state the tuning; as I understand it, chords synthesised from notation in this way are in *equal temperament*, the standard keyboard tuning in which every semitone is the same size, and this paper treats them so. Only the mean rating of each chord is available.

Chord ratings of Bowling et al. (2018). Bowling et al. (2018) collected consonance ratings for all 298 chords formed by a lowest note and one to three further notes, each a whole number of semitones above the lowest note and within one octave of it: 12 dyads, 66 triads and 220 tetrachords. The 30 listeners were 15 students of a music conservatory in Singapore and 15 students recruited at the University of Vienna; the Vienna group had, on average, less than a year of weekly music lessons. They rated each chord on a four-point scale. The article describes the scale as running from 1 ("quite consonant") to 4 ("quite dissonant"), but its reported average ratings and the data files I used are coded so that a higher value means more consonant: the documentation of Harrison's data package `inconData` states that "high ratings indicated high consonance" (Harrison, 2019), and the octave has the highest average. This paper reads the data in that way; for the chord regressions alone the ratings are then turned round, so that a higher value is more dissonant (Method section). The stimuli were piano tones in *just intonation*, in which intervals are exact whole-number frequency ratios: the tritone was 7/5 (582.5 cents) and the minor sixth 8/5 (813.7 cents). These are the frequencies of Bowling et al.'s supplementary materials, as distributed in `inconData` (Harrison, 2019). Again only the mean rating of each

chord is available. Both chord sets were obtained from the public collection of consonance rating data sets of Digital and Cognitive Musicology Lab, EPFL (2025) and cross-checked against Harrison (2019).

Why dense dyads. The chord sets contain one rating at the tritone and one at the minor sixth, together with a small number of other scale intervals, and those two intervals otherwise occur inside chords with other notes. Marjieh et al. (2024) point out that stimuli drawn only from a scale leave out whatever structure lies between the scale steps. For the present question the practical consequence is that a curve cannot be fitted to "all other intervals" from twelve points. The dense dyad ratings supply the missing intervals, and they are the main data of this paper.

Dense dyad ratings of Marjieh et al. (2024). According to the article's statement of author contributions, the experiments were designed by Raja Marjieh, Peter Harrison and Nori Jacoby. All five authors list the Max Planck Institute for Empirical Aesthetics in Frankfurt am Main among their affiliations (four of them together with a second institution), and the experiments ran online with PsyNet, the authors' software for online experiments (Marjieh et al., 2024). Participants in the United States were recruited through Amazon Mechanical Turk, a website on which people do short tasks for pay. A second cohort in South Korea, native Korean speakers resident there, was recruited locally with the help of a research assistant; the article's Methods section gives the reason, that Mechanical Turk did not run in South Korea at the time. Marjieh et al. added this cohort to test whether the effects they found are specific to US listeners. These are the only two cohorts in the released data, which is why the present analysis covers these two countries and no others. Participants had to pass a short listening test that checks that headphones are being worn.

On each trial a participant heard one dyad and rated how well the word "pleasant" matched the sound, from 1 (completely disagree) to 7 (completely agree). The interval was drawn from a uniform distribution between 0 and 15 semitones, and the lower tone from a range of ten semitones (G3 to F4 in note names, that is, the five semitones on either side of middle C, C4, about 262 Hz), so each participant heard a different random set of dyads and, in principle, each stimulus was rated once. In the released files each participant contributed on average 39 to 107 ratings, depending on the experiment. Marjieh et al. define musicians as participants with more than two years of musical experience and non-musicians as those with two or fewer; I use the same split.

Each experiment used one tone spectrum. I used every dyad rating experiment with a fixed spectrum: harmonic complex tones of ten harmonics whose strength falls by 3 decibels (dB) per octave; the same tones with partials *stretched*, so that the second partial lies at 2.1 times the frequency of the first and not at twice, or *compressed* to 1.9 (partial n lies at $2.1^{\log_2 n}$ or $1.9^{\log_2 n}$ times the first, where a harmonic tone has $2^{\log_2 n} = n$; figure labels give this value as "octave 2.1" or "octave 1.9"); five equal-amplitude harmonics with and without the third harmonic; pure tones; a tone imitating the bonang, an instrument of the Javanese gamelan (a percussion orchestra) whose partials are not harmonic, paired with a harmonic lower tone; and tones built from the measured harmonic strengths of a flute, a guitar and a piano (*resynthesised* tones). Six of these ten US spectra are harmonic complex tones: the ten-harmonic tones, the two tones of five harmonics, flute, guitar and piano. I call them the harmonic spectra; the other four are the stretched, compressed, pure and bonang tones. The Korean cohort rated the ten-harmonic, stretched and compressed stimuli. The released rating files are complete in ratings (for the ten-harmonic tones they hold the 7,500 ratings listed in the article's Supplementary Table 1), but

the number of participant identifiers in them differs from the number of participants the article reports. The files of the ten US experiments contain 147 to 194 identifiers each, where Marjieh et al. report 149 to 210 participants (for the ten-harmonic tones, 165 against 198). The files of the three Korean experiments contain 37 to 45 identifiers each, more than the 20 to 24 participants reported. The article's Methods section states that each Korean participant was allowed to take part in the same experiment up to three times, which would account for the larger number if an identifier stands for one sitting and not for one person; the files do not say which identifiers belong to the same person. For the smaller numbers in the US files I found no explanation in the article, its Supplementary Information or the repository. I analysed the files as released and treat each identifier as one participant: ratings are standardised within an identifier, and the resampling behind the confidence intervals (Method section) redraws identifiers. I made one exclusion of my own. A participant who gave the same rating on every trial has no spread of ratings, so the standardisation described in the Method section cannot be computed, and such participants were dropped: at most 6 in a US experiment, 33 in all, and none in the Korean experiments. This leaves 146 to 191 participants in each US experiment (1,627 participant records in total, a record being one participant in one experiment) and 37 to 45 in each of the three Korean experiments (122 records). Supplementary Table S2 gives the three counts for every experiment. The 13 experiments as analysed comprise 95,865 ratings. Participant identifiers are local to each experiment, so these totals count a person once for every experiment taken, and whether US participants took part in more than one experiment cannot be told from the files. I treat experiments as independent samples; to the extent that they share listeners, the uncertainty of results combined over experiments is somewhat understated (Section 10). A further experiment, the roll-off experiment (Study 3 in the numbering of Marjieh et al., 2024), varied the *roll-off*, the rate at which the harmonics become weaker, from trial to trial within one group of participants (322 in the article, 301 in the released file, 295 after the exclusion above); I analyse it at three levels of my choosing (2, 7 and 12 dB/octave; the Method section says how the ratings at one level are obtained) and, for one exploratory analysis, at five levels from 1 to 13 dB/octave. With it the total is 110,565 ratings.

What was used and what was not. A data set was used if its ratings are public, the sounds are isolated (not part of a melody or chord sequence), and the notes of each sound are known exactly. From Marjieh et al.'s release I used only experiments in which listeners rated dyads; their ratings of three-note chords are used only for one check in the Supplementary Information. I did not search systematically for further chord rating sets (Section 10).

Access. Ratings and the synthesis parameters of every spectrum were taken from the authors' public repository (Harrison, 2023). The repository holds no audio; where a model needs the partials of a stimulus I computed them from the released synthesis parameters. For the flute, guitar and piano the available spectrum is a single set of harmonic strengths from one analysed tone. I used it for every note, which is an approximation, since the spectrum of a real instrument changes from note to note; I therefore call these three *proxy* spectra. The code that produces every number in this paper, and the instructions for obtaining the data, are in the repository named under Code availability.

5 Models

All models in this section were proposed by others. Each takes the partials of a sound (their frequencies and strengths) and returns one number; computing it for every interval gives a curve

with the interval on the horizontal axis. Formulas, constants and the checks of my implementations are in Supplementary Section S1.

Roughness. The Hutchinson–Knopoff model (Hutchinson & Knopoff, 1978), written H&K from here on, takes every pair of partials, measures their separation relative to the critical bandwidth, and assigns the pair a roughness that is zero when the partials coincide, largest at about a quarter of a critical bandwidth, where beating is most unpleasant, and zero again when they are far apart. The pair's roughness is weighted by the strengths of the two partials and the results are added. A high value means a rough sound, so the curve is high for intervals predicted to be unpleasant. I also report the model of Sethares (1993), which has the same structure with a different formula for the roughness of a pair, and the revised H&K model of Marjieh et al. (2024), which adds a term for slow beating and adjusts how strength is weighted; its constants were fitted by Marjieh et al. to the dense dyad ratings.

Harmonicity. I implemented the harmonicity model of Harrison and Pearce (2018, 2020). In plain terms, the model asks, for every candidate pitch, how well the partials of the sound fit the harmonics of a note at that pitch. A sound whose partials fit one candidate much better than the rest, like a single note, has high harmonicity; a sound whose partials fit no candidate well, or many equally, has low harmonicity. A high value therefore means a sound predicted to be pleasant, the opposite direction to roughness. A partial rarely falls exactly on a harmonic, so the model needs a *tolerance* (written σ in tables and figures): how far, in cents, a partial may lie from a harmonic and still count as fitting it. (Technically it is the width of a bell-shaped weighting, so the fit falls off gradually and not at a hard limit.) The model as published by Harrison and Pearce has a tolerance of 6.83 cents, which I call the published tolerance. The main result of this paper uses 20 cents instead, which accepts slightly mistuned partials. That value is not taken from the literature. The code released with Marjieh et al.'s paper defines, but does not use, a variant with a 15-cent tolerance (Harrison, 2023). The list of values I tried (6.83, 10, 15 and 20 cents) is mine and was written down before the screen was run, and the data selected 20 cents from it: each time one experiment was set aside, the value that fitted the remaining experiments best was taken, and it was 20 cents every time. The experiment set aside was then used only to judge the resulting curve. I call this comparison of candidate curves on experiments left out of the fit the screen (Method section, "The screen"). The values 6.83 and 20 cents are the only model constants that matter for reading the results. The model has one further setting, how strongly each harmonic of the candidate note counts, which differs slightly between its two published uses. I did not tune it: for dyads I use the setting of Marjieh et al. (2024) and for chords that of `incon` (Harrison, 2024), a software collection of consonance models by Harrison.

The composite. Marjieh et al. (2024) combine their revised roughness model and the harmonicity model with fixed weights. I call this the *composite*. Because it was fitted to the same dense dyad ratings, its agreement with them is partly *in-sample*: a model always fits the data it was tuned on better than it fits new data.

Familiarity and periodicity (chords only). Two further models are used only in the chord analysis of Section 7.7, as computed by `incon`. The familiarity model of Harrison and Pearce (2020) counts how often each chord, regardless of the note on which it is built, occurs in a collection of transcribed popular music (Burgoyne et al., 2011); a frequent chord is predicted to be more pleasant. The periodicity model of Stolzenburg (2015) measures how long the combined

sound wave of a chord takes to repeat; a short repetition period is predicted to be more pleasant. Periodicity is a different property from beating, which belongs to roughness. Neither model is applied to the dense dyads, and this paper does not settle the role of familiarity: no data here separate what listeners have learned from what the ear does (see Section 9.2).

6 Method

Input and output (this study). The input of every analysis is listeners' ratings together with a model curve. The output is a difference between *average ratings*, or between an average rating and the value a curve predicts for it. No sound wave or spectrum is ever subtracted from another: the models turn sounds into numbers, and every comparison is then made between numbers on the rating scale. Two orders are fixed throughout: the rating at the minor sixth minus the rating at the tritone, and the observed rating minus the value a curve predicts. Any other difference is written out where it is used.

From ratings to a profile (method of Marjieh et al.). Following Marjieh et al. (2024), I first converted each participant's ratings to *z-scores*: the participant's own mean was subtracted and the result divided by the participant's own standard deviation (SD). This removes differences in how listeners use the scale. All effects in this paper are therefore in units of a listener's own rating SD; averaged over the six US experiments with harmonic tones, one SD corresponds to 1.24 points on the seven-point scale, so an effect of +0.20 SD is about a quarter of a scale point. The z-scores were then averaged as a function of interval by *kernel smoothing*: the value at an interval is a weighted mean of the ratings at nearby intervals, with weights that fall off as a Gaussian (bell-shaped) function of distance, of SD 0.2 semitones as in Marjieh et al. The result, evaluated every 0.01 semitones, is the *pleasantness profile* $P(c)$: the average rating as a function of the interval c in semitones. A high value means that listeners found the interval pleasant. Each model was evaluated for the stimulus of every trial, and its output was averaged over interval in the same way as the ratings; this smoothed model curve is what is later fitted to the profile. The z-scoring and the smoothing width are those of Marjieh et al.; the grid, the name "profile" and the averaging of model output over the trials actually presented are my implementation choices. In the roll-off experiment the ratings were weighted in the same manner along the roll-off axis as well (Gaussian, SD 1.5 dB/octave, the width in Marjieh et al.'s code), to obtain a profile at each chosen roll-off. The conversion of SD units to scale points above is my own computation.

The gap and the dip (this study). The *gap* is $P(8) - P(6)$: the average rating at the minor sixth minus the average rating at the tritone. A positive gap means that the minor sixth is rated as more pleasant. The *dip* is $\frac{1}{2}[P(5) + P(7)] - P(6)$: the mean rating of the fourth and the fifth minus the rating of the tritone. A positive dip means that the tritone is rated below its two neighbours.

Fitting a curve and the residual (this study). Model output is in arbitrary units, so it must be put on the rating scale before it can be compared with ratings. I did this by *least squares*: I chose a constant (the *intercept*) and one weight per model so that the weighted model output plus the intercept lies as close as possible to the profile, in the sense of the smallest sum of squared differences. The fit used the intervals from 0.5 to 14.75 semitones, except those within ±0.6 semitones of the tritone and of the minor sixth; I call each excluded stretch a *window*. I

call the result the *fitted curve*. The weight of roughness comes out negative, which turns the roughness curve upside down: on the fitted curve, as on the profile, higher always means more pleasant. Because the tritone and the minor sixth are left out, the fitted curve at those two intervals is a prediction from the rest of the interval axis. The *residual* $r(c)$ is the observed profile minus the fitted curve at interval c; a negative residual means that listeners rated the interval below the curve's prediction. The *unexplained gap* is $r(8) - r(6)$, the part of the gap that the curve does not predict. A curve that predicted the gap exactly would leave zero. The dip left unexplained by a curve is defined in the same way, $\frac{1}{2}[r(5) + r(7)] - r(6)$. I call a negative residual at the tritone the tritone's *shortfall*: listeners rate the tritone lower than the curve predicts. Wherever a value is given it is the residual itself, with its sign.

In-sample and held-out fits (this study). In descriptive results the intercept and weights are fitted to the experiment being described. Such a fit is *in-sample*. For the main results the weights come from other data: for each US experiment in turn, the weights were fitted to the nine remaining US experiments, whatever their spectrum, and applied to the one left out (*leave one experiment out*; each such round is called a fold in the tables). The nine are fitted together in a single least-squares fit, with the two windows excluded in each: their profiles are placed one after another, each keeps an intercept of its own, and all nine share one weight per model. The profiles have the same number of points, so every experiment counts equally. In the left-out experiment only the intercept, which absorbs its general rating level and does not change the shape of the curve, is fitted, over the same intervals as before (the two windows excluded). I call these fits *held-out*. A Korean experiment is predicted in the same way with weights fitted to all ten US experiments. The model output is always computed for the stimuli of the experiment being predicted. Several fits are used in this paper; Table 3 lists which analysis uses which.

The screen (this study). The screen asks whether any further property of the stimulus improves the curve of H&K roughness plus harmonicity. The baseline for every comparison is that curve with the published tolerance. Before running it, I wrote its rules at the top of the analysis script: a list of candidate additions, all computed from the stimulus and none tied to a particular interval, and the criterion for comparing them: how well a curve predicts the whole profile of an experiment that was left out of the fit, measured by R^2, the share of the variance of that profile that the curve accounts for (1 for a perfect prediction), averaged over the left-out experiments. The unexplained gap and dip of the left-out experiments were recorded alongside. The candidates were as follows. (1) A wider tolerance of mistuning in the harmonicity model. The tolerance was chosen from 6.83, 10, 15 or 20 cents anew each time an experiment was left out, as the value with the smallest sum of squared differences between fitted curve and profile over the remaining experiments. The left-out experiment therefore played no part in the choice. (2) Root ambiguity, an idea of Terhardt (1974) and Parncutt (1988); "root" here means the virtual pitch of the Introduction, the single pitch that the ear assigns to the whole sound. The harmonicity model gives every candidate pitch a score for how well the partials fit it, and I computed root ambiguity from two scores, the best one and the highest among the candidate pitches at least one semitone away from it: the mean score over all candidate pitches is subtracted from both, and the second is then divided by the first. (3) Temporal periodicity, the largest normalised autocorrelation of the summed partials over lags of 2–30 ms, that is, how strongly the sound wave repeats itself within that time. This is my own implementation of an idea found in Pressnitzer et al. (2001), Stolzenburg (2015) and Masina et al. (2022), and is not the periodicity model of the Models section, which is used only for chords. (4) As a deliberate negative control, harmonic entropy, a measure proposed by Paul Erlich (Tonalsoft, n.d.). It

is computed from the frequency ratio of the two notes alone. The ratio is treated as heard imprecisely, so that it could be taken for any of several nearby ratios of small whole numbers, each with some probability; harmonic entropy is the entropy of those probabilities, high where no single simple ratio stands out. It varies smoothly with the interval and ignores the spectrum, so it cannot respond to the differences between the tones of the experiments. It is a candidate that should not help if the screen works. The formula for (2) and the code for (2) to (4) are mine. Candidates (2) to (4) were each added on their own to roughness plus harmonicity with the published tolerance. One further curve contained (2) and (3) together, with the tolerance chosen as in (1). All weights were refitted each time an experiment was left out. These rules were written before the screen was run, but after the in-sample results of Sections 7.2 and 7.3 and the stretched-spectrum comparison had been seen. They were not deposited in a public registry of study plans (the study was not pre-registered), so the statement rests on my own files.

Primary estimand (this study). An *estimand* is the quantity an analysis sets out to estimate. The question, how much of the gap a dissonance curve leaves, was set before the dense ratings were analysed, and the rules of the screen, which selects the curve, before the screen was run. In the screen the tolerance came out as 20 cents every time an experiment was left out, and no other candidate improved the prediction (Section 7.4). The selected curve is therefore H&K roughness plus harmonicity with a tolerance of 20 cents, and wherever the 20-cent curve is used after the screen the tolerance is fixed at that value, not chosen again. Which single number is reported as the answer was decided later, after results had been seen (Section 10). It is the held-out unexplained gap $r(8) - r(6)$ under the selected curve. The weights are fitted to the nine other US experiments, and the gap is computed in each of the 6 US experiments with harmonic spectra (the ten-harmonic tones, the two tones of five harmonics, flute, guitar and piano) and then pooled over them as described below. The pooling is restricted to these six because the question concerns ordinary harmonic tones; the pure, bonang, stretched and compressed tones are reported separately. Every other estimate is secondary; those found after seeing the data are labelled exploratory where they appear.

Why this design (this study). *Why the minor sixth.* A roughness curve gives the tritone and the minor sixth nearly the same value (Figure 1), so any difference between their ratings is a difference that roughness does not produce; the minor sixth is the natural reference for the tritone. *Why the curve is fitted to the other intervals.* The model's units are arbitrary, so a discrepancy can only be defined relative to a fit, and fitting the curve to the two intervals under test would let it absorb part of the effect. *Why the dip uses the fourth and the fifth.* They are the tritone's neighbours, one semitone to either side, so the dip measures how far the tritone lies below its surroundings. *Why held-out fits.* A curve fitted to an experiment can follow the noise of that experiment; weights from other experiments cannot. *Why one primary estimand.* With many models, spectra and intervals there are many numbers to choose from, and choosing after seeing them would bias the result. The question and the rules of the screen were written down before the corresponding analyses were run, but the choice of this one number was not, which is a limitation (Section 10). *The window.* The ±0.6 semitones excluded around each interval is my choice: it is three times the width of the smoothing kernel (0.2 semitones), the distance beyond which a rating contributes almost nothing to the smoothed value at the interval. As an exploratory check I repeated the primary analysis with ±0.4 and ±0.8 semitones; the primary estimand did not change to two decimal places (+0.18 and +0.18 SD, against +0.18 with ±0.6; Supplementary Section S2).

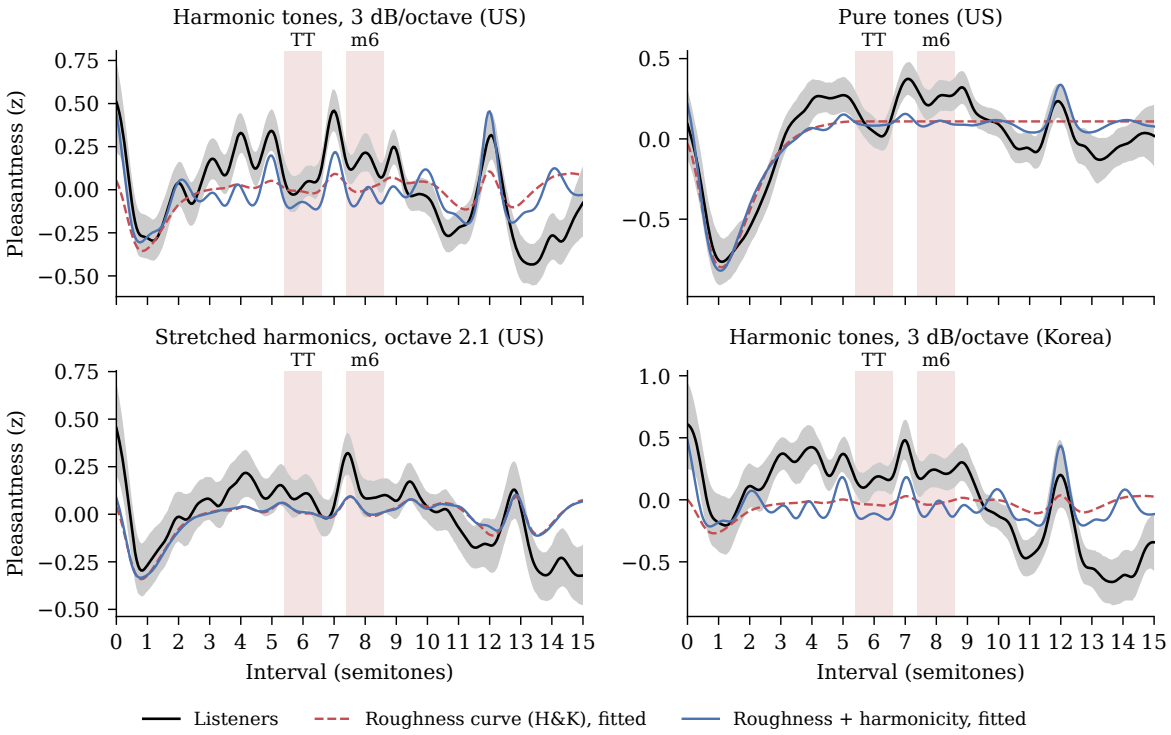


Figure 1: *This study.* Listener pleasantness profiles (black; the grey band is the 95% confidence interval, the range of values compatible with the data, defined under "Uncertainty and pooling") with the roughness curve (H&K, red dashed) and the roughness-plus-harmonicity curve (blue, published tolerance), each placed on the rating scale of the experiment shown (in-sample) by a least-squares fit over the intervals from 0.5 to 14.75 semitones outside the shaded windows at the tritone (TT) and minor sixth (m6); the Method section gives the details. Horizontal axis: interval in semitones. Vertical axis: average rating in units of a listener's own rating standard deviation; higher means more pleasant. Ten-harmonic tones (top left); pure tones (top right); stretched partials (bottom left); Korean listeners, ten-harmonic tones (bottom right). Where the two fitted curves coincide, the blue one hides the red one.

Uncertainty and pooling (standard methods, applied here). A 95% *confidence interval* (CI) is the range of values compatible with the data; it is given in square brackets after each estimate. CIs come from the *bootstrap*: participants were redrawn at random with replacement 1,000 times and the whole analysis was repeated on each draw, so the spread of the results shows how much an estimate depends on who took part. Results from several experiments were combined by *random-effects (RE) meta-analysis*, a weighted average that allows the true effect to differ between experiments, using the estimate of the between-experiment variance by DerSimonian and Laird (1986). The CI of the pooled value is the Hartung–Knapp interval, which allows for the uncertainty of the between-experiment variance and is therefore wider and more trustworthy when experiments are few (Hartung & Knapp, 2001; IntHout et al., 2014). I report τ, the SD of the true effect between experiments; I^2, the share of the observed variation that is due to real differences between experiments and not to chance (Higgins & Thompson, 2002); and a 95% *prediction interval*, the range in which the effect in a new experiment would be expected to fall. Confidence intervals from the DerSimonian–Laird method alone are in Supplementary Table S4; of the 326 pooled estimates stored in the results files of this study (all models, quantities and groups of experiments), 22 differ between the two methods in whether their CI excludes zero.

Every interval (this study). The primary estimand says how large the unexplained gap is. This secondary analysis asks a different question: where on the interval axis the gap arises, and whether the tritone is unusual among the intervals. It does not replace the primary estimand. The held-out residual was computed at every whole-number interval from 1 to 14 semitones, from the held-out fits described above (windows excluded only at the tritone and the minor sixth), once with the published tolerance and once with the 20-cent tolerance. Because the weights come from other experiments, the shape of the curve owes nothing to the experiment being predicted, at any interval. Only its constant is fitted there, and the other intervals, unlike the tritone and the minor sixth, contributed to that constant. In each experiment two differences were then formed, and each was pooled over the harmonic spectra. The first is the tritone's residual minus the mean residual of the ten other intervals from the minor second to the major seventh (1 to 11 semitones, that is, the intervals inside the octave, the minor sixth included); a negative value means that the tritone lies further below the curve than the other intervals do. The second is the minor sixth's residual minus the mean residual of five intervals traditionally classed as consonant (minor and major third, fourth, fifth, major sixth). Because the residual is relative to an intercept fitted on the rest of the interval axis, it shows where the shape of the curve fails, not how pleasant an interval is. The residual is defined on the whole interval axis from 0 to 15 semitones, including the points not used in the fit.

The conclusions about single intervals rest on these two differences. Section 7.5 also reports a second comparison, which asks the same question with dips. For every whole-number interval k from 1 to 14 semitones a separate curve of roughness plus harmonicity (published tolerance) was fitted in-sample with a single window, placed at k, and the dip left unexplained at k, $\frac{1}{2}[r(k-1)+r(k+1)]-r(k)$, was taken from it. The quantity reported is the tritone's value minus the mean of the other 13, pooled once over the six harmonic spectra and once over those six with the pure and bonang tones.

Stretched and compressed spectra (this study, exploratory). The stretched and compressed tones are those of Marjieh et al.; using them to locate the tritone's low rating is my analysis. In these tones the two notes of a dyad are as far apart as before, but their partials no longer coincide where those of harmonic tones do. If the low rating belongs to the interval of six semitones as such, the dip should stay at 6 semitones. If it follows the partials, it should move to the interval at which the partials of the two notes relate as they do at the tritone of harmonic tones. With the second partial at s times the first ($s = 2.1$ or 1.9 in place of 2), that interval is $6\log_2 s$ semitones, 6.42 for the stretched and 5.56 for the compressed tones; I call it the tritone of the *stretched scale* (a name I use for the compressed tones as well), and the other intervals of that scale lie at $k\log_2 s$ semitones in the same way.

Each of these experiments was therefore analysed twice, with two separate fits. The first is the analysis used for every other experiment: tritone at 6 semitones, minor sixth at 8, fourth and fifth at 5 and 7, and the two windows at 6 and 8. The second uses the stretched scale throughout: tritone at $6\log_2 s$, minor sixth at $8\log_2 s$, fourth and fifth at $5\log_2 s$ and $7\log_2 s$, and the two windows at the stretched scale's tritone and minor sixth. Each analysis gives four numbers by the formulas of the paragraphs above: the observed gap and dip of the profile P, and the gap and dip left unexplained by the curve of roughness plus harmonicity (published tolerance), fitted in-sample. The paired difference is one of these four numbers from the second analysis minus the same number from the first, with a CI from the same bootstrap draws. A positive paired difference in the observed dip means a deeper dip at the stretched scale's tritone than at 6 semitones, which is what is expected if the low rating follows the partials. Separately from these in-sample fits, the screen described earlier also recorded, for each candidate, the held-out

dip left unexplained at 6 semitones (neighbours at 5 and 7) and at the stretched scale's tritone (neighbours at $5\log_2 s$ and $7\log_2 s$) for the US stretched experiment, both from the screen's one held-out fit (windows at 6 and 8) (Table 6).

Targeted tests (this study). Four differences (contrasts) probe what the unexplained part depends on. The stimuli and the groups of listeners are those of Marjieh et al.; using them as tests of the tritone is my analysis. All four use curves fitted to each experiment separately (in-sample) and, where harmonicity enters, the published tolerance, as they were first computed. (1) Culture: the unexplained gap after roughness and harmonicity for Korean listeners minus that for US listeners, for the same ten-harmonic tones. (2) Training: the same unexplained gap for musicians minus that for non-musicians, each from a curve fitted to that group's own profile; the difference is formed in each experiment and then pooled over the eight US experiments without stretched or compressed partials. (3) Third harmonic: by my reading of the roughness model, the third harmonic creates the roughness that distinguishes the fourth and fifth from the tritone, so removing it should change the dip. The dip of a fitted curve F is computed like that of the profile, $\frac{1}{2}[F(5)+F(7)]-F(6)$, with H&K roughness alone fitted to each experiment. The contrast is the change in the observed dip (five harmonics without the third minus five harmonics with it) minus the same change in the dip of the fitted roughness curve. (4) Roll-off: the change in the observed gap (2 dB/octave minus 12 dB/octave) minus the same change in the gap of the fitted roughness curve, $F(8)-F(6)$. A separate curve is fitted to the profile at each roll-off level. Contrasts (3) and (4) use roughness alone because they test a prediction of the roughness model itself. The four were among the first comparisons computed for the dense ratings (Supplementary Table S26 reports the others). My files do not record that they were singled out before their results were seen, and treating them as one family (a set of tests judged together) was decided afterwards. They are corrected by *Holm's procedure* (Holm, 1979), which raises the bar for each test so that the chance of any false positive among the four stays at 5%.

Chords (this study). For the chords I ran one regression for each set of models in Table 8: none (chord size only), H&K roughness, Sethares roughness, and H&K roughness with harmonicity, with familiarity, with both, or with periodicity. Each regression predicts the rating from chord-size indicators (variables that mark whether a chord has two, three or four notes), the outputs of the models in the set, the number of tritones in the chord (pairs of notes whose distance is six semitones after whole octaves are subtracted, counted from the chord's keyboard steps and not from its exact tuned frequencies) and, as a further control, the number of pairs one semitone apart, counted in the same way. The coefficient of the tritone count is the *tritone penalty* left unexplained (rating units; CIs from redrawing chords with replacement, since only the mean rating of each chord is available). For the chord analyses only, all ratings are oriented so that a higher value is more dissonant (the Johnson-Laird ratings are published in that direction, and I reverse the Bowling ratings), so a positive penalty means that a chord is rated as more dissonant, for each tritone it contains, than the models predict. For the twelve Bowling dyads I report, for each model, a *share*: the tritone minus the minor sixth on that model's curve fitted to the twelve dyads, divided by the same difference in the ratings, in per cent (a ratio, so the orientation of the ratings does not affect it). A share of 100% means that the model reproduces the whole difference. With only twelve points the two intervals cannot be left out of the fit, so the share is descriptive.

The tuning check announced earlier is a comparison of two runs. For the Johnson-Laird chords, treated as equal-tempered (Data section), the model outputs are those of `incon`. For the Bowling chords every regression and every share was computed twice: once with `incon`'s

outputs for the chords entered as whole numbers of semitones, that is, in equal temperament, and once with roughness and harmonicity recomputed by my code at the just-intoned pitches that were presented. The familiarity and periodicity outputs are `incon`'s in both runs: periodicity was not recomputed at the presented tuning, and familiarity is defined on keyboard pitch categories (which keys are pressed, whatever their exact tuning) and does not depend on the tuning. These regressions use the published tolerance; Section 7.7 also reports the penalty with the 20-cent tolerance.

When each analysis was fixed (this study). Table 3 lists, for each analysis, the kind of fit, the tolerance, the experiments and when it was fixed. Nothing in this study was registered publicly, so its last column rests on my own files. The first analyses run were the chord regressions and, for the dense ratings, the gap and dip in every experiment and the differences between groups of listeners and between spectra. The screen came next: its rules, with the list of candidates, were written at the top of its script before it was run, but after those first results had been seen. Where this paper says that something was fixed beforehand, it means only that it was written down before that analysis was run. The table marks what was chosen after results had been seen, and the text marks those analyses as exploratory where they are reported.

Noise ceiling (this study, standard method). Ratings are noisy, so even a perfect model could not match a profile exactly. The *noise ceiling* is the best agreement that the reliability of the data allows. I estimated it by *split-half reliability*: the participants of an experiment were split at random into two halves (50 splits), a profile was computed from each half, and the two were correlated. The Spearman–Brown formula (Spearman, 1910; Brown, 1910) then gives the reliability of the full sample from that of the halves. Two measures of agreement are used in this paper and should not be confused. The correlation coefficient, written r without an argument (the residual is always written $r(c)$), runs from -1 to 1 and measures how closely two curves have the same shape. R^2 is the share of the variance of a profile that a curve accounts for; it is at most 1 and can fall below 0 for a curve that predicts a left-out experiment worse than a constant would. The reliability given by the Spearman–Brown formula is itself the noise ceiling. It estimates the share of the variance of the observed profile that is not noise, and that share is the R^2 that a perfect, noise-free model would be expected to reach against the observed profile.

How correctness was checked (this study). *Models.* For each of the 13 experiments I compared the curve from my implementation, smoothed, with the smoothed model curve that Marjieh et al. (2024) published for the same stimuli. The Pearson correlation r between the two curves was at least 0.9999 for H&K roughness, 0.9996 for harmonicity and 0.9999 for revised H&K. These values show that my code reproduces the published models; they say nothing about how well a model fits listeners, which is measured by R^2. On the Bowling chords my implementations give the same values as `incon` when given the same equal-tempered pitches. *Numbers.* No result of my analysis in the text or tables was typed by hand: each is written by the analysis scripts to a results file and inserted from there, and a script (`check_consistency.py`) verifies that every cross-reference in the text points to an existing table, figure or section and that the abstract, Introduction and Conclusion contain no hand-typed result. *Reproducibility.* Starting from an empty directory, downloading the public data and rerunning the whole pipeline reproduced every stored result to within an absolute difference of 10^{-9}.

7 Results: main analyses

This section reports the main analyses; the last column of Table 3 says when each was fixed. Several parts of it were fixed only after results had been seen: the choice of the primary estimand, the paired comparison of the two curves (Section 7.3), the residual at every interval (Section 7.5), the grouping of the four targeted tests and the stretched-spectrum test (both in Section 7.6). For the comparison of dips across intervals (Section 7.5) the time is not recorded. They are kept here because they answer the questions of this section. The other analyses chosen after seeing the data are in Section 8. Everything reported is my analysis of the published ratings, and where a sentence reports another study's result it names that study.

How to read the numbers. Unless stated otherwise, every value is in units of a listener's own rating SD, where one SD is 1.24 points on the seven-point scale for harmonic tones, and is followed by its 95% CI in square brackets. The sign convention is the same throughout. A positive *gap* means that the minor sixth is rated above the tritone. A positive *unexplained gap* means that listeners separate the two intervals by more than the fitted curve does; zero would mean that the curve predicts the gap in full. A negative *residual* at an interval means that listeners rated it below the curve. In all tables bold type marks a value whose CI excludes zero. To keep this section short, it gives the results combined over experiments and the tables needed to follow the argument; the results for every single experiment are in Supplementary Section S16.

7.1 Where the tritone stands among the intervals (prior data and this study)

Among the twelve Bowling dyads (Table 4), 4 intervals are rated less consonant than the tritone (1.953), but the minor sixth is rated higher (2.477). The dense data show the same. In Figure 1 the horizontal axis is the interval in semitones and the vertical axis the average rating in SD units. The black line is the listeners' profile and the coloured lines are fitted curves, so where the black line lies above a coloured line, listeners rated that interval higher than the curve predicts. The tritone sits in a local trough, well above the minor second and major seventh. That the tritone is "the most dissonant interval" describes its role in tonal syntax more than how an isolated dyad sounds (Parncutt & Hair, 2011). The narrower claim, that it is heard as less pleasant than a dissonance curve predicts, is the one tested here.

7.2 Roughness alone (this study)

Combined over the six US experiments with harmonic spectra, the minor sixth is rated +0.26 SD [+0.16, +0.36] above the tritone, and the gap left by fitted H&K roughness is +0.26 SD [+0.15, +0.37]: roughness predicts none of the observed gap. The Sethares model gives the same result (Supplementary Tables S22 and S23 list every experiment and every model). The reason can be seen in Figure 1. Between the fourth and the fifth the roughness curve is nearly flat, so once fitted it predicts almost the same pleasantness at the tritone and the minor sixth. Roughness does separate the tritone from the fourth and fifth, whose low partials coincide, but its fitted curve reproduces only a small part of the tritone's dip below them (the pooled dip left unexplained is +0.25 of the +0.30 observed). For pure tones, whose partials do not interact at these intervals, roughness predicts no gap, yet listeners produce one (+0.21 SD [+0.08, +0.34]).

7.3 Adding harmonicity (this study)

The primary estimand is the gap that remains when the curve of roughness plus harmonicity (20-cent tolerance) is applied to experiments it was not fitted to. It is +0.18 SD [+0.04, +0.32] (τ = 0.12, I^2 = 75%), where roughness alone, in the same held-out fits, leaves +0.26. Harmonicity therefore removes part of the gap, and the larger part remains. The remaining gap is real on average but differs between spectra: the prediction interval for a new harmonic spectrum, [−0.18, +0.55], includes zero. The synthetic spectra leave little (ten-harmonic tones +0.12, five equal harmonics +0.07), and the resynthesised flute, guitar and piano leave most (+0.29, +0.36 and +0.27). Because the spectra available for those three instruments are proxies, this contrast may reflect spectral detail that the models do not see as much as a property of the listeners.

The descriptive fits, in which each curve is fitted to the experiment itself with the published tolerance, agree (Figure 2: the black point is the observed gap or dip and the coloured points are what each curve leaves unexplained, so a coloured point at zero means that the curve accounts for the whole effect). Harmonicity without roughness leaves less of the gap but fits the rest of the interval axis worse, and the composite of Marjieh et al. (2024), whose parameters were fitted to these experiments, also leaves part of the gap (Supplementary Table S23).

Which end of the gap? A gap is a difference, $r(8) - r(6)$, so it can arise at either end. Table 5 splits it. Under roughness alone the tritone lies −0.09 [−0.22, +0.03] SD from its prediction and the minor sixth +0.17 [+0.06, +0.29]; relative to the mean residual of the other intervals from the minor second to the major seventh, the tritone is −0.18 [−0.30, −0.05] below. The tritone's shortfall under roughness is therefore relative, not reliably negative in absolute terms. With harmonicity (20-cent tolerance) the tritone lies at −0.02 [−0.15, +0.11] and its difference from the other intervals is −0.11 [−0.25, +0.02], neither distinguishable from zero, while the minor sixth stays at +0.17 [+0.03, +0.31].

A residual that is distinguishable from zero under one curve and not under the other does not show that the two curves differ, so I also computed the change between the curves directly, after the results above had been seen (exploratory; Supplementary Table S3). Adding harmonicity changes the residual at the tritone by +0.07 [+0.06, +0.08], the residual at the minor sixth by 0.00 [−0.04, +0.04], and the gap by −0.07 [−0.11, −0.04]. These intervals are narrow, and that should not be over-read. The ratings are the same under both curves, so each interval describes only how consistently the two curves differ in their predictions when the listeners are resampled. It does not say how precisely the ratings are known; the wider intervals of Table 5 do that. The values in this section are rounded separately, so they need not add up exactly, and the change in the gap should be read from its own interval. The reading is that harmonicity lowers the predicted pleasantness of the tritone slightly and consistently and does not detectably change the prediction for the minor sixth, so the part of the gap it removes is at the tritone. It does not turn a detectable shortfall at the tritone into none, because under roughness alone the tritone's own residual was already not distinguishable from zero; only its position relative to the other intervals was. What is left of the gap is the minor sixth rated above the curve. The minor sixth is not unusual in this: its residual does not differ from the mean of the minor and major thirds, fourth, fifth and major sixth (+0.02 [−0.15, +0.18]; Section 7.5). These residuals are relative to an intercept fitted on the rest of the interval axis, so they locate where the shape of the curve is wrong, not absolute levels.

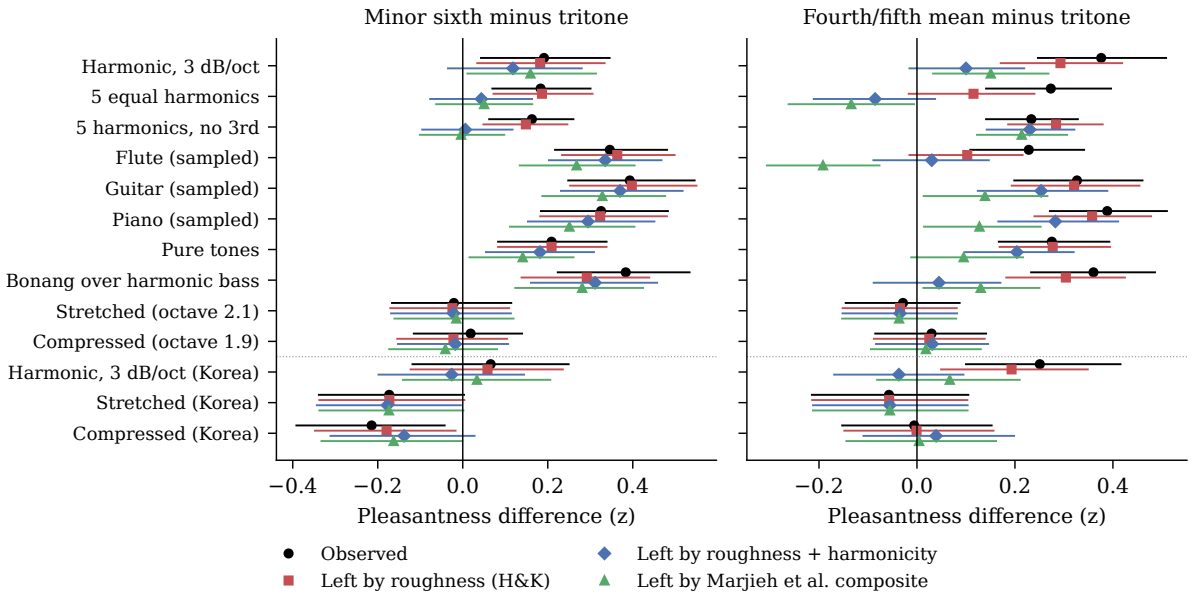


Figure 2: *This study.* Observed gap and dip (black) and the part left unexplained by roughness (red), roughness plus harmonicity (blue) and the fitted composite of Marjieh et al. (2024) (green), per experiment, each model fitted to the experiment itself, with 95% bootstrap CIs. Below the dotted line: stretched and compressed spectra and the Korean cohort.

7.4 Held-out prediction and the screen (this study)

With one pair of weights fitted across US experiments (published tolerance) and each experiment predicted from weights fitted to the others, the held-out unexplained gap is positive in 8 of 10 experiments and its CI excludes zero in 5; for harmonic spectra, in 6 and 3 of 6 (Supplementary Table S24). The residual on unseen spectra is in the same range as the descriptive one, so it is not an artefact of refitting each experiment.

The screen. Only the tolerance helped (Table 6, in which each row is one candidate curve and a higher R^2 means better prediction of experiments not used for fitting; Figure 3). The tolerance chosen was 20 cents every time an experiment was left out, and the two-term curve then reaches a mean held-out R^2 of 0.548, against 0.533 with the published tolerance (paired difference +0.015 [+0.005, +0.023]) and 0.493 for the composite of Marjieh et al. (2024) (paired difference +0.055 [+0.032, +0.079]). The composite's internal weights were fitted to these experiments, so the comparison favours it. Root ambiguity and periodicity added nothing once harmonicity was present, and the negative control added nothing, as intended. The wider tolerance improves the curve as a whole but barely moves the gap, and no candidate removed it.

Two cautions apply. First, 20 cents is a representative value, not an estimate of a property of hearing. Checked afterwards (exploratory), the best tolerance depends on how strongly the ratings are smoothed: 10 cents with a kernel of 0.1 semitones and 20 with 0.3. Before running this check I had set the criterion that the best tolerance should lie within 15–25 cents at both kernels; it was met at 0.3 but not at 0.1 semitones. A tolerance wider than the published one helps at every kernel width. Second, with the dyads sampled continuously, a wider tolerance means only that listeners' consonance peaks are broader than the model's; it is not evidence about equal temperament as such (Section 7.7). Rows examined after the screen are in Supplementary Table S6.

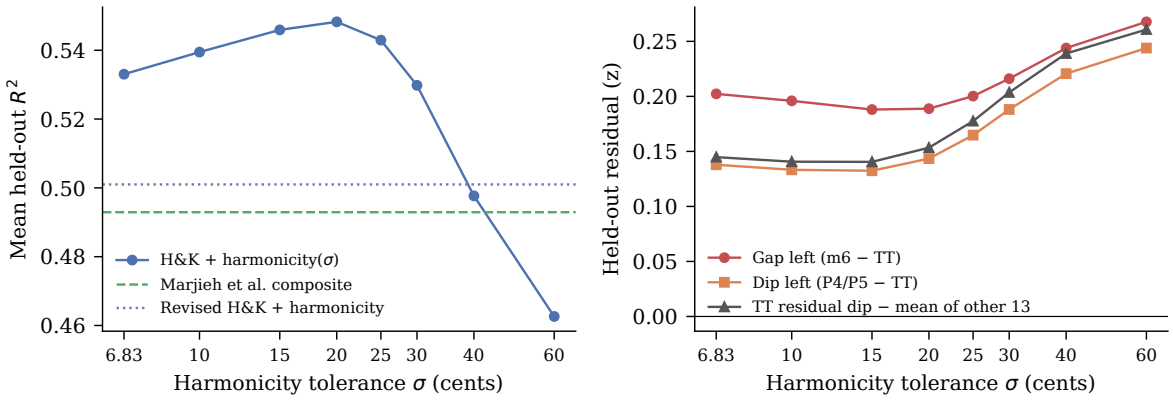


Figure 3: *This study.* Harmonicity tolerance. Left: mean held-out R^2 of roughness plus harmonicity as the Gaussian width σ varies (published value 6.83 cents), against revised roughness plus harmonicity and the composite of Marjieh et al. (2024). Right: the tritone residuals left on held-out harmonic experiments. Values beyond 20 cents were examined after the screen.

7.5 The residual at every interval (this study)

This analysis was added after the screen, to locate the unexplained gap. The held-out residual was computed at every whole-number interval and pooled by random effects over the six harmonic spectra (Supplementary Table S25). Under roughness alone the most negative residuals between the minor second and the major seventh are those of the major second (−0.10) and the tritone (−0.09 [−0.22, +0.03]). After harmonicity with the 20-cent tolerance the consonant intervals (thirds, fourth, fifth and sixths) lie above the curve, while the seconds, the tritone and the minor seventh lie near it (tritone −0.02 [−0.15, +0.11]). The tritone is thus one of several intervals that the curve predicts about right, not an interval detectably rated below it.

One comparison points the other way, and I report it rather than drop it. The same question can be asked with dips instead of residuals (Method, last paragraph of "Every interval"). Measured in this way, the tritone's dip after harmonicity (published tolerance) differs from that of the average interval by +0.16 [−0.01, +0.32] over the six harmonic spectra, an interval that includes zero, but by +0.16 [+0.04, +0.27] when pure tones and the bonang are added. Since the tritone's own residual is not below average, that excess reflects the fourth and fifth rated above the curve on either side, not a shortfall at the tritone. Per-interval dips are shown in Supplementary Figure S1.

7.6 Stretched spectra and four targeted tests (this study)

Spectrum (exploratory). The comparison at the stretched scale's tritone was set up after the first results for the dyads had been seen, so it is exploratory. With stretched or compressed partials the unexplained gap at 600 cents is about zero (US, after roughness and harmonicity: −0.02 and −0.02 SD), and so is the observed dip (−0.03 and +0.03), although the profiles are not flat. A penalty attached to the physical 600-cent interval would survive stretching, since the fundamentals are still 600 cents apart. With the second partial at s times the first, the scale that matches the spectrum has its tritone at $6 \log_2 s$ semitones (Method): 6.42 for the stretched and 5.56 for the compressed tones. There the stretched tones show an observed dip of +0.19 [+0.07, +0.31] SD, where there was none at 6 (paired difference +0.22 [+0.07, +0.37]). The negative control, attached to the frequency ratio, necessarily puts its penalty at 600 cents instead. For compressed tones the dip at 5.56 is not significant (+0.07 [−0.05, +0.19]), and the

Korean listeners show no clear dip at either position, so the shift rests on one spectrum and one cohort.

Targeted tests. Four contrasts probe what the residual after roughness and harmonicity depends on (Table 7): Korean against US listeners, musicians against non-musicians, tones with and without the third harmonic, and a shallow against a steep roll-off. They were computed early; their choice as a family of four, and the Holm correction over them, came afterwards (Method). After Holm correction 0 of the four remain significant (adjusted p from .180 to .746), so the data do not establish that the unexplained part depends on any of the four.

Uncorrected, each contrast has a direction (group values in Supplementary Table S26). Musicians show a larger observed gap and dip, part of which follows from harmonicity, to which musicians are more sensitive (McDermott et al., 2010); after roughness and harmonicity the difference includes zero (gap +0.10 [−0.03, +0.23]), and the residual is present in non-musicians. Korean listeners show a smaller gap for harmonic tones, but they differ from US listeners in the same direction for stretched and compressed partials, where US listeners show no gap, so the difference is not specific to consonance. Removing the third harmonic changes the observed dip less than roughness predicts, consistent with Marjieh et al. (2024), who found that the third harmonic matters through harmonicity. In the roll-off experiment the gap does not shrink as the upper partials, and with them the roughness, fade (Supplementary Table S10).

7.7 Chords (this study, prior data)

Scored in equal temperament, the twelve Bowling dyads show the pattern of the dense data (Table 8): roughness reproduces −2% of the tritone–minor-sixth difference and roughness plus harmonicity 48%. In the chords, each tritone costs +0.080 [+0.006, +0.150] rating units after roughness and harmonicity. But the chords were played in just intonation. At the presented tuning, harmonicity rates the minor sixth (8/5) higher than in equal temperament, while the tritone (7/5) barely changes. Scored as presented, roughness plus harmonicity reproduces 93% of the dyad difference and the chord-level tritone penalty falls to +0.009 [−0.056, +0.074]. With twelve dyads the share is an overfit-prone statistic; the chord-level penalty, estimated from 117 chords containing a tritone, is the firmer result.

For the Johnson-Laird chords, which I take to be equal-tempered (Data section), a penalty remains with the published tolerance: +0.39 [+0.08, +0.69] rating units per tritone after roughness and harmonicity (the mean ratings of these chords have an SD of 1.10). Those chords were rated on a seven-point and not a four-point scale, have their notes spread over a wider range of pitch and were rated by a different population, so the two sets, and the sizes of their penalties, are not directly comparable. The 20-cent tolerance (the wider harmonicity setting selected in the screen) reduces the Johnson-Laird penalty to +0.28 [−0.08, +0.60], no longer distinguishable from zero; the reduction itself is distinguishable from zero in the paired bootstrap. For the just-intoned Bowling chords the same tolerance explains less of the variance in the ratings than the published tolerance does (both comparisons are in the released file `results/chords_sigma.json`). One possible reading is that the tolerance that suits the ratings depends on the tuning listeners hear, but the two chord sets differ in other ways as well, so the comparison does not establish it. Further chord results are in Supplementary Section S14.

7.8 The curves against the noise ceiling (this study)

The split-half ceilings range from 0.82 to 0.97 (US mean 0.93). Held out, the recommended curve captures 59% of the explainable variance in the US experiments and the composite of

Marjieh et al. (2024) 53%. The shortfall is not at the tritone. Fitted to single experiments, the residuals share a broad shape: intervals in the middle of the range are rated above the curve and wide intervals, above all beyond the octave, below it. A linear term in interval size removes most of this (in-sample R^2 0.38 to 0.73 for harmonic tones). Its slope depends on the spectrum (Section 8.3). Interval size is confounded here with the pitch of the upper tone and with sharpness, the weighting of a spectrum towards high frequencies (Fastl & Zwicker, 2007). Sharpness was a component of sensory pleasantness in psychoacoustics (Aures, 1985) before consonance models narrowed to roughness and harmonicity, it lowers the consonance of chords in high registers (Eerola & Lahdelma, 2022), and for pure-tone dyads preference falls as mean frequency rises (Kaklamani & Simserides, 2026).

8 Exploratory analyses

The analyses in this section were not planned in advance: each was chosen after seeing the data, so each can suggest a hypothesis but cannot confirm one. They are all my analyses of the published ratings (this study), and none of them changes the primary result of Section 7.

8.1 A bonus for in-tune intervals and a step

Two further patterns were found after seeing the residual profile. Both are computed on the held-out residual of the curve with the 20-cent tolerance and a linear term in interval size (which absorbs the smooth trend of Section 7.8). The first is a local bonus: the residual at an equal-tempered interval c minus the mean at the quarter-tones on either side, $b(c) = r(c) - [r(c - 0.5) + r(c + 0.5)]/2$. Pooled over the harmonic spectra, the intervals from the minor second to the major seventh other than the tritone have a mean bonus of +0.14 [+0.08, +0.19] SD and the tritone +0.07 [0.00, +0.14]. The difference between the two, +0.07 [−0.01, +0.14], includes zero. What the data do separate, and what I noticed only after seeing the profile, is that the major second and minor seventh receive a smaller bonus than the other eight intervals (−0.12 [−0.14, −0.10]); the tritone does not differ from them (+0.03 [−0.04, +0.09]). The minor second and the major seventh, which are rougher, receive the full bonus, and I do not know what separates the three intervals with a smaller one.

The bonus belongs to harmonic tones: it is +0.14 [+0.12, +0.15] for harmonic spectra but +0.04 [+0.01, +0.07] for pure tones and +0.04 [0.00, +0.07] for the bonang, and at either scale of the stretched and compressed tones it is not distinguishable from zero (next paragraph). At a smoothing width of 0.2 semitones it cannot tell equal temperament from just intonation, so I read it as a bonus for the neighbourhood of the familiar intervals, not for their exact tuning. Marjieh et al. (2024) noted that details of the timbral effects remain unexplained by their models; I am not aware that this bonus has been quantified.

The second pattern is a step. Writing $r(c) = b(c) + q(c)$, with $q(c)$ the mean residual at the two neighbouring quarter-tones, the baseline q rises between 7 and 8 semitones by +0.10 [−0.03, +0.23] (an indicator after a linear trend). The step is absent for the ten-harmonic tones (−0.02 [−0.12, +0.07]) and the flute (+0.01 [−0.08, +0.10]), and large for the guitar (+0.31 [+0.22, +0.41]) and piano (+0.17 [+0.08, +0.27]). The guitar and piano spectra are proxies, one analysed tone each, so the step may come from the spectral model rather than from the listeners. I therefore do not decompose the gap into a fixed share of step and bonus; the decomposition for each spectrum is in Supplementary Table S27.

The bonus under stretched spectra. The in-tune bonus does not follow the stretched scale. At the stretched scale's own intervals the mean bonus is +0.02 [−0.02, +0.05] (compressed +0.01 [−0.02, +0.05]), against +0.13 [+0.10, +0.17] for harmonic tones, and the minimum effect these data could detect with 80% power is 0.05 SD (Supplementary Table S9; exploratory). The Korean listeners show the same pattern (harmonic +0.07 [+0.03, +0.12], stretched −0.04 [−0.08, +0.01]). The dip moves with the partials; the bonus stays with harmonic tones.

The published harmonicity model cannot follow the shifted dip, because its template is a harmonic series. Replacing the template with the upper tone's own partials, one reading of the proposal that harmonic templates are learned from the sounds heard (Shamma & Klein, 2000), reproduces the stretched-scale dip but predicts harmonic tones worse, so it is not a better general curve (exploratory; Supplementary Section S9).

8.2 Tuning of the minor-sixth peak

Located with a narrower kernel (SD 0.1 semitones; exploratory), the listeners' minor-sixth peak lies at 808 [796, 819] cents, between equal temperament and 8/5 (813.7 cents) and −6 [−17, +6] cents from the harmonicity model's peak, and the gap left is no smaller at just-intoned positions (difference +0.03 [−0.11, +0.17]), so these data cannot tell whether listeners heard the minor sixth as tempered or just (inconclusive; Supplementary Section S13).

8.3 The smooth trend and sharpness

The largest shortfall of the curves is the smooth trend over interval size described in Section 7.8. Figure 4 shows its slope in each experiment (panel A; a negative slope means that the wider the interval, the further listeners rate it below the curve), the same slope at each roll-off (panel B), and the held-out R^2 with and without a sharpness term against the noise ceiling (panel C). As a comparison chosen after seeing the residuals, I added one smooth term with a weight shared by all experiments. Held-out US R^2 rises from 0.548 to 0.625 with interval size and to 0.658 with a sharpness proxy computed from the synthesis parameters, that is from 59% to 71% of the explainable variance; widening the tolerance, by comparison, added +0.015. The gain is concentrated in the synthetic spectra, and two spectra do not fit it: for five equal harmonics the trend is absent although sharpness rises with the interval, and the flute, whose ratings rise with interval size, is predicted worse with the sharpness term. In the roll-off experiment the slope over interval size flattens as the roll-off steepens, from −0.45 [−0.60, −0.33] at 1 dB/octave to −0.13 [−0.22, −0.03] at 13, as sharpness predicts and interval size alone does not; but the proxy's implied slopes cover only 50% of that flattening, and I had looked at this experiment before choosing sharpness.

One test was set up in advance, and it failed. For the one rating set of Marjieh et al. (2024) not yet analysed, 80 triads of five equal harmonics rated by 192 participants, I wrote down three predictions before reading the data: a negative sharpness weight, a better leave-one-chord-out R^2, and a better correlation of the frozen-weight curve with the chord means. None passed (Supplementary Section S12), although the triads leave little room for a third term. I therefore treat the smooth trend as established and sharpness as a candidate explanation, and I do not add it to the recommended curve.

Korean listeners and the smooth trend. The models fit the Korean profiles poorly (roughness plus harmonicity, in-sample $R^2 = 0.14$ for harmonic tones), although their ceilings are high (0.946–0.950). Predicted with weights from US listeners, the recommended curve does worse than each experiment's mean (held-out R^2 −0.19 [−0.26, −0.08] on average); adding a linear

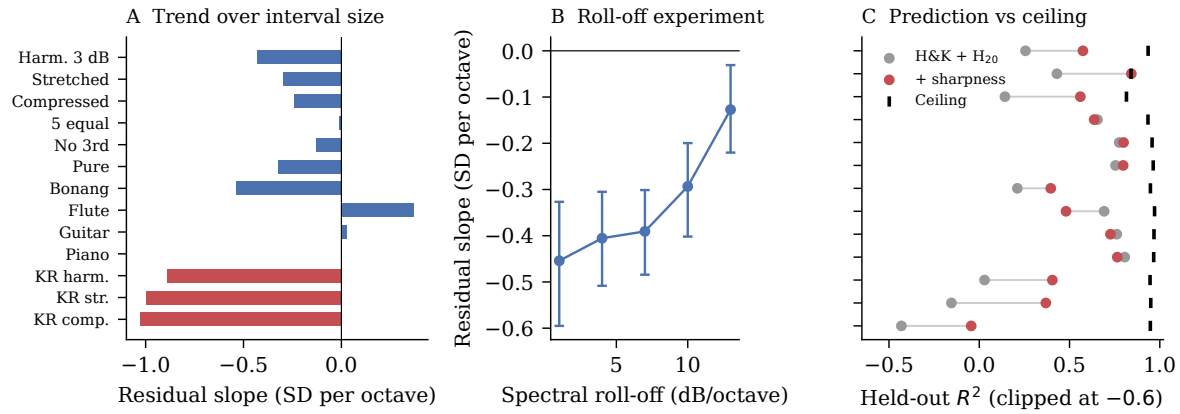


Figure 4: *This study.* The smooth trend (exploratory). (A) Slope over interval size of the residual from H&K roughness plus harmonicity (20-cent tolerance), each experiment fitted alone (SD per octave of interval). (B) The same slope in the roll-off experiment at each spectral roll-off (participant-bootstrap 95% CI). (C) Held-out R^2 per experiment without and with the sharpness term, against the split-half ceiling (Korean experiments predicted with weights from all US experiments).

term in interval size brings it to +0.11 [−0.01, +0.22] and a sharpness proxy to +0.24 [+0.09, +0.36](Supplementary Table S11; exploratory). What separates the cohorts in these data is thus mainly a steeper decline of pleasantness with interval size, not a different ordering of consonances. Its cause is open: sharpness, which rises with interval size, is one candidate, and in an online study so is the playback equipment, which may differ between cohorts.

9 Discussion

9.1 What closes part of the gap, and the recommended curve

For harmonic tones and US listeners the minor sixth is rated +0.26 within-listener SD above the tritone, and a roughness curve fitted to the rest of the interval axis predicts none of it: roughness is almost flat between the fourth and the fifth. The popular diagnosis, that the dissonance curve is an oversimplification, is correct in a precise sense. Part of what is missing is harmonicity. The tritone's partials fit no single harmonic series well, whereas those of the minor sixth approximate the series whose 5th and 8th harmonics are the two fundamentals. Adding harmonicity lowers the curve at the tritone slightly, leaves it unchanged at the minor sixth, and so removes a small part of the gap (Section 7.3). Root ambiguity, periodicity and harmonic entropy add nothing further: an interval whose partials fit no single series is the same interval whose virtual pitch is ambiguous.

The curve I recommend is $P(x) = \alpha_e + b_R\, R_{\mathrm{HK}}(x) + b_H\, H_{20}(x)$, with R_{HK} the Hutchinson–Knopoff roughness of the dyad's combined spectrum, H_{20} Harrison and Pearce's harmonicity computed with a 20-cent tolerance (11-harmonic template weighted $1/i$, the same Gaussian width for spectrum and template, lower tone at C4), both smoothed with the 0.2-semitone kernel, and pooled weights $b_R = -1.65$ and $b_H = 0.66$ for the 20-cent curve (per unit of each raw model output; α_e is an experiment intercept). It has no interval-specific term. It predicts held-out US experiments better than the published combinations I tested, but only slightly better than the same curve with the published tolerance; for just-intoned tones the published tolerance fits better. Neither the curve family nor its ingredients are new (Parncutt et al., 2019; Harrison & Pearce, 2020; Marjieh et al., 2024; Masina et al., 2022); what is new is that its tritone

predictions were tested on experiments not used to fit it, against candidates listed before that comparison was run, and on a spectrum that moves the intervals.

Three cautions apply to anyone who uses it. First, its weights are for new spectra of the kind tested here; on the experiments themselves they were refitted without the experiment being predicted, and the intercept makes the curve a statement about the shape of the profile, not its level. Second, the tolerance is a smoothing choice as much as a psychoacoustic parameter, since it interacts with the kernel that smooths ratings and model outputs alike; at a finer resolution the 20-cent curve smooths away the minor-sixth peak that listeners show (Supplementary Section S13). Third, the curve is meant for equal-tempered tones; chords presented in just intonation should be scored as presented and with the published tolerance. A smooth term over interval size would improve it more than anything tested at the tritone, but until its cause is known I leave it out.

9.2 What remains, and how to read it

What remains in dyads is the primary estimand, +0.18 SD [+0.04, +0.32], about 0.24 points on a seven-point scale for the in-sample fit. It is not a constant in three respects. It varies between spectra more than sampling error allows (prediction interval [−0.18, +0.55]), with the largest values for the proxy spectra of the sampled instruments. It is absent at 600 cents when the partials are stretched or compressed. And it is not detectably the tritone's: after harmonicity the tritone's residual does not differ from zero (under roughness alone it did not reliably do so either), while the minor sixth lies above the curve, like the other consonances. The exploratory decomposition does not change this. Its step is carried by two proxy spectra, and the tritone's reduced in-tune bonus does not separate it from the other intervals once spectra are treated as a random sample; the one contrast that is clear, a smaller bonus for the major second and minor seventh, was found after seeing the profile. On these data I would not add a tritone penalty to a dissonance model for dyads. That does not extend to chords: in the chords of Johnson-Laird et al., taken here to be equal-tempered, a penalty for each tritone remains after roughness and harmonicity with the published tolerance, although with the 20-cent tolerance it is no longer distinguishable from zero (Section 7.7). Harmonicity is worth adding for the part of the gap it removes, but the larger task is to find what makes consonant intervals of harmonic tones more pleasant than any current curve predicts.

Pure tones. Pure tones might seem to cut the other way: they have no upper partials to align, yet they leave one of the largest gaps. The residual profile resolves this. For pure tones the tritone is predicted about right (−0.01) and the gap is the minor sixth (+0.18), so pure tones give no evidence of a penalty at 600 cents in the first place. Because their partials are the fundamentals, pure tones cannot separate spectral accounts from accounts based on the frequency ratio; only a spectrum whose partials depart from the harmonic series can.

The stretched tones and the heard octave. With stretched partials the dip leaves 600 cents and appears at the stretched scale's tritone (an exploratory result, Section 7.6). This could be read as compatible with a learned category defined relative to the octave the listener actually hears. That reading predicts more than the dip: a category re-centred on the heard octave should also carry the in-tune bonus to the stretched scale's intervals. It does not (+0.02 [−0.02, +0.05], with a detectable effect of 0.05, against +0.13 [+0.10, +0.17] for harmonic tones). The dip follows the partials and the bonus does not, which favours a spectral account of the dip, in which partials of the two tones line up or fail to, over a re-centred learned category. This fits the

finding that consonance judgements for inharmonic bell timbres follow their partials rather than the nominal intervals (Harrison & MacConnachie, 2024). It rests on one stretched spectrum; the compressed tones show no clear dip at their own tritone.

Two readings of the minor-sixth residual. The first is that the harmonicity model is incomplete. The tuning analysis does not identify the defect: the listeners' minor-sixth peak lies between equal temperament and just intonation, and the gap is no smaller at just-intoned positions. A harmonicity model wrong in some other detail would have to explain why the residual varies between spectra, why it is shared by the other consonant intervals, why it vanishes in the just-intoned Bowling chords and why a penalty persists in the Johnson-Laird chords, taken here to be equal-tempered. The second reading is learned: familiarity with the consonant intervals of tonal music, heard mostly in harmonic sounds (Lahdelma et al., 2022; McDermott et al., 2016). The in-tune bonus fits this reading, since it appears for harmonic tones and not at either scale of the stretched and compressed ones. But the targeted tests do not support its usual corollaries once corrected for multiplicity: neither training (compare McDermott et al., 2010; Popescu et al., 2019) nor culture changes the residual detectably. These two readings are those of the current debate, between psychoacoustic accounts in which harmonicity is a perceptual mechanism (Harrison & Pearce, 2020; Marjieh et al., 2024) and the view that harmonicity partly stands in for exposure (Lahdelma et al., 2022; Milne et al., 2023). These data locate the residual at consonant intervals of harmonic tones; they cannot tell which reading is right.

The chords sharpen the contrast without resolving it. The just-intoned Bowling chords leave no tritone penalty after roughness and harmonicity, and the Johnson-Laird chords (taken to be equal-tempered) leave one, which the 20-cent tolerance reduces to a value whose interval includes zero ($+0.28$ $[-0.08, +0.60]$). This is what an incomplete harmonicity model would produce if it under-credits equal-tempered consonances, and it is also what a learned preference for familiar equal-tempered intervals would produce. The two chord sets differ in voicing and population as well as tuning, and in chords the tritone penalty is estimated from a count of tritones, so the chords do not show that the tritone is special either; they show that tuning must be scored as presented.

What would decide. No analysis of the present data can separate the two readings; the experiments that could are listed under Further research in the Conclusion.

9.3 What dissonance curves can say

Roughness curves computed from measured instrument spectra do not settle the ordering of the tritone and the minor sixth. Across eight instruments at C4 and four kernels, the instrument accounts for 55% of the variance of the tritone/minor-sixth roughness difference, the kernel for 8% and their interaction for 36%; the rarely reported shape constants of the H&K kernel move it by a median 13.9 pp, and one clarinet spectrum reverses the listeners' ordering under H&K in two libraries (Supplementary Section S15). Combined with the finding that listeners do not follow roughness in this region even for idealised spectra, roughness curves should not be used to argue about the tritone either way. I recommend that analyses report the kernel and its constants, the partial-merging and peak-acceptance rules, the recording and register, the tuning at which the stimuli were presented, and whether the conclusion holds for a second recording; and that any new curve be judged on held-out spectra and against the noise ceiling, not against the previous curve.

10 Limitations

Data. No new listener data were collected. The dense data are online pleasantness ratings of synthetic or resynthesised dyads heard in isolation; the tritone's reputation concerns musical context, which these data do not reach. Only two cohorts exist in the released data, and the Korean one is small, differs from the US cohort in recruitment and probably in playback equipment as well as in culture, and its profiles are poorly fitted. Experiments are treated as independent, although participants may have taken part in more than one; the overlap is unknown because identifiers are local to each experiment. Shared listeners would make intervals pooled over experiments somewhat too confident. The three Korean files hold 122 identifiers where Marjieh et al. (2024) report 68 participants (Data section). The article states that a Korean participant could take the same experiment up to three times, so some identifiers probably belong to the same person, and the intervals for the Korean experiments, and for the comparison of cohorts, are then too narrow. The musicianship split is observational, with a coarse two-year threshold. I did not search systematically for other public rating sets, so the choice of the two chord sets reflects what was available with exact stimulus descriptions, not a complete survey.

Models. Roughness here is a model quantity, not a measured percept (McBride, 2025). For the flute, guitar and piano the spectra are proxies from one analysed tone, and the largest residuals, and the step of the exploratory decomposition, are found there. Models were evaluated with the lower tone at C4, whereas it was sampled from G3 to F4; averaging roughness over that range changes the held-out R^2 of the recommended curve by -0.001 [-0.001, 0.000] and its dip by $+0.001$ [$+0.001$, $+0.001$]. The 20-cent tolerance is a representative value and depends on the smoothing; it is not an estimate of a property of hearing.

Estimands and statistics. Nothing was registered publicly, and the statements about what was fixed beforehand rest on my own files. The rules of the screen were written down before it was run, but after the first in-sample results had been seen; the primary estimand was chosen later still, after results had been seen; and the four targeted tests were chosen as a family after they had been computed, so the Holm correction does not allow for that choice. The paired comparison of the two curves at the tritone (Section 7.3) was added last. The window excluded from the fit is my choice; the result did not change with windows of ± 0.4 or ± 0.8 semitones, but only those two alternatives were tried. Six harmonic experiments are few for estimating heterogeneity, which is large; Hartung–Knapp intervals are wider than DerSimonian–Laird intervals, and for 22 of 326 pooled estimates the two methods differ in whether the interval excludes zero. The four targeted tests are corrected for multiplicity; the many other contrasts are not. A residual that does not differ from zero is not evidence that it is zero: the intervals show how large an effect at the tritone the data still allow.

Exploratory analyses. The residual profile, the paired comparison of the two curves, the test at the stretched scale's tritone, the in-tune bonus, the step, the stretched-scale bonus, the tuning analysis, the sharpness term, the Korean held-out fits and the window check are exploratory. The stretched-scale results rest on one spectrum and one cohort, and the test of the heard-octave account assumes that a re-centred category would carry the bonus as well as the dip. The tuning analysis could not separate equal temperament from just intonation: the peak's interval spans both, and equivalence was not shown (Supplementary Section S13). The sharpness proxy is computed from synthesis parameters, was chosen after seeing the residuals and failed the triad test, whose predictions were written down beforehand (not publicly registered).

Chord analyses. The Bowling analysis rests on 298 chords, twelve of them dyads, rated by 30 listeners, and the Johnson-Laird chords differ from them in rating scale, voicing (how widely the notes are spread) and population and, on my inference, in tuning: that the Johnson-Laird chords were in equal temperament is inferred from how they were synthesised, and the article does not state their tuning (Data section). Only mean ratings per chord are public, so the chord bootstraps resample chords, not raters. In chords the tritone penalty is estimated from a count of tritones that lie mostly inside chords with other notes, which is a weaker measurement than the dyad gap.

11 Conclusion

What was asked and what was done. This paper asked why listeners prefer the minor sixth to the tritone when the standard dissonance curve, built on roughness, rates the two about the same. It answered by reanalysing published ratings, not by running a new experiment. The difference between the two intervals was compared with a curve fitted to all the other intervals, and the curves were judged on experiments they had not been fitted to.

What was found. First, the premise holds and roughness does not explain it: for harmonic tones US listeners rate the minor sixth +0.26 within-listener SD above the tritone, and a roughness curve predicts none of that difference. Second, harmonicity explains a small part of it. Adding harmonicity to the curve, in a version that tolerates more mistuning than the published one, lowers the prediction at the tritone and leaves a gap of +0.18 SD [+0.04, +0.32]. Third, what remains is not at the tritone. It sits at the minor sixth, which listeners rate above the curve as they do the other consonant intervals, and it varies between spectra; the tritone's own rating does not lie reliably below either curve. Fourth, the tritone is not where the curves fail most: their largest shortfall against the reliability of the ratings is a smooth trend over interval size.

What did not work. Several things I tried did not give the result I was looking for, and they belong to the record as much as the results above. In the screen, the further properties of the sound that I tested, root ambiguity and periodicity, did not improve the curve. Only the wider tolerance did, and it improved the curve as a whole while barely changing the gap. The value of that tolerance is itself not firm: the best value depends on how strongly the ratings are smoothed, and the criterion I had set for its stability was met at one smoothing width and not at the other. None of the tests of training, culture, third harmonic or roll-off survives correction for multiple testing, so the data do not show what the remaining gap depends on. The shift of the tritone's dip with stretched partials, the most direct sign of an acoustic origin, appears for one spectrum and one cohort: it was not found with compressed partials or in the Korean listeners, and the analysis was set up after the first results had been seen. The curves fit the Korean ratings poorly. Sharpness, my candidate explanation for the smooth trend, failed all the predictions that I wrote down before testing it on a further rating set. Finally, the single number reported as the main result was chosen after results had been seen, not before.

What the results do not show. These conclusions are limited to isolated two-note sounds rated online by two cohorts, and they do not show that the tritone has no penalty of its own, only that these dyad data do not detect one. In the chords of Johnson-Laird et al., taken here to be equal-tempered, a penalty for each tritone remains under the published harmonicity

model, though not reliably under the 20-cent one (Section 7.7). They also leave the cause of the remaining gap open: an incomplete harmonicity model and a learned familiarity with consonant intervals both fit what was found.

What could be done next. A next study of this kind should state its question, its main quantity and its tests in a public registry before the data are analysed, which this one did not. Beyond that, four studies would take the question further. A replication of the stretched-spectrum result with a second stretch factor and a second cohort would test whether the dip follows the partials in general. A dense sampling of the regions around the minor sixth and the major third, at a resolution finer than the 14 cents that separate the two tunings and with enough listeners per spectrum, would locate the listeners' peaks well enough to show whether the harmonicity model misplaces them. Listeners with little exposure to equal-tempered harmonic music, rating the same dense dyads, would test the learned reading directly; the Korean cohort here is too small and too different in recruitment to do so. Finally, the gap and the dip should be measured in musical context, where the tritone's reputation was formed: isolated dyads show how a tritone sounds, not how it functions.

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Data availability

No new data were collected; the study reanalyses published data. The dense dyad ratings and timbre parameters are from Harrison (2023) (`https://gitlab.com/pmcharrison/timbre-and-consonance-paper`), the chord ratings from Digital and Cognitive Musicology Lab, EPFL (2025) (`https://github.com/DCMLab/consonance`) and Harrison (2019) (`https://github.com/pmcharrison/inconData`), and the instrument recordings from Philharmonia Orchestra (n.d.) (`https://philharmonia.co.uk/resources/sound-samples/`) and University of Iowa Electronic Music Studios (n.d.) (`https://theremin.music.uiowa.edu/MIS.html`). The code repository provides scripts that fetch these data rather than copies; the partials extracted from the recordings are included as derived data.

Code availability

All analysis code, intermediate results and the manuscript source are available at `https://github.com/boenchen1112/tritone-harmonicity-dyads` (release `v1.0-submission`; archived at `https://doi.org/10.5281/zenodo.23272654`). Every number in the text and tables is generated from JSON result files by `analysis/make_tables.py`, and every figure by `analysis/make_figures.py`; the exceptions are design constants and values quoted from the literature.

Ethics statement

This is a secondary analysis of published, anonymised data. No new data were collected from human participants, and no ethical approval was required.

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Competing interests

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Table 3: *This study.* Overview of the analyses: which fit and which tolerance of the harmonicity model each one uses, on which experiments, and when it was fixed (see "When each analysis was fixed"). In-sample: curve fitted to the experiment itself; held-out: weights fitted to the other US experiments.

Analysis	Fit	Tolerance (cents)	Experiments	When fixed
Gap and dip, per experiment and pooled (Supplementary Tables S22 and S23)	in-sample	6.83	all 13; pooled over the 6 US with harmonic spectra, and over those 6 with pure tones and bonang	first analysis of the dense ratings
Gap and dip with held-out fits (Supplementary Table S24)	held-out	6.83	10 US; Korean predicted from all US	with the screen
Screen (Table 6)	held-out	chosen from the list in each round	10 US; Korean predicted from all US	rules written before it was run, after the first results
Primary estimand (Table 5)	held-out	20, fixed	6 US with harmonic spectra	this number chosen after results had been seen
Residual at every interval (Supplementary Table S25)	held-out	6.83 and 20, each fixed	6 US with harmonic spectra	added after the screen
Dip compared across intervals (last paragraph of "Every interval"; Section 7.5)	in-sample	6.83	6 US with harmonic spectra; 8 with pure tones and bonang	when it was chosen is not recorded
Stretched and compressed spectra (Section 7.6)	in-sample, two fits	6.83	stretched and compressed, US and Korean	after the first dyad results were seen (exploratory)
Targeted tests (Table 7)	in-sample	6.83	as named in each test	computed early; chosen as a family of four afterwards
Chords (Table 8)	regression over chords	6.83; 20 also reported	two chord sets	first analysis of the chords
Exploratory analyses (Section 8)	stated with each analysis			after seeing the data

Table 4: *Prior data.* Mean consonance ratings for the 12 dyads of Bowling et al. (2018) (higher = more consonant; the minor and major sevenths are tied).

Interval	Semitones	Rating	Rank (1 = least consonant)
Minor second	1	1.317	1
Minor seventh	10	1.753	2
Major seventh	11	1.753	3
Major second	2	1.837	4
Tritone	**6**	**1.953**	**5**
Minor sixth	**8**	**2.477**	**6**
Minor third	3	2.693	7
Major sixth	9	2.713	8
Perfect fourth	5	2.950	9
Major third	4	3.173	10
Perfect fifth	7	3.447	11
Octave	12	3.893	12

Table 5: *This study.* Which end of the gap. Held-out residuals (observed − predicted) at the tritone, $r(6)$, and the minor sixth, $r(8)$, under H&K roughness and under the recommended curve (H&K + harmonicity, 20-cent tolerance), and the gap left by the latter, in SD [participant-bootstrap CI]. Pooled row: random effects over the six harmonic spectra, Hartung–Knapp CI; its last cell is the primary estimand. Bold: CI excludes zero.

	Roughness (H&K)		H&K + harmonicity (20 cents)		
Experiment	$r(6)$	$r(8)$	$r(6)$	$r(8)$	Gap left
Harmonic, 3 dB/oct	+0.03 [−0.07, +0.12]	**+0.21 [+0.09, +0.32]**	+0.10 [0.00, +0.19]	**+0.22 [+0.10, +0.33]**	+0.12 [−0.03, +0.27]
5 equal harmonics	+0.07 [−0.03, +0.17]	**+0.25 [+0.18, +0.33]**	**+0.15 [+0.04, +0.25]**	**+0.22 [+0.14, +0.29]**	+0.07 [−0.03, +0.19]
5 harmonics, no 3rd	**−0.16 [−0.24, −0.08]**	−0.02 [−0.10, +0.07]	**−0.11 [−0.19, −0.02]**	−0.08 [−0.17, 0.00]	+0.03 [−0.07, +0.13]
Flute	−0.07 [−0.18, +0.03]	**+0.29 [+0.18, +0.39]**	+0.02 [−0.09, +0.12]	**+0.31 [+0.20, +0.40]**	**+0.29 [+0.15, +0.42]**
Guitar	**−0.23 [−0.34, −0.12]**	**+0.16 [+0.04, +0.29]**	**−0.16 [−0.27, −0.04]**	**+0.20 [+0.08, +0.32]**	**+0.36 [+0.20, +0.52]**
Piano	**−0.18 [−0.28, −0.09]**	**+0.14 [+0.04, +0.26]**	**−0.10 [−0.20, 0.00]**	**+0.17 [+0.06, +0.28]**	**+0.27 [+0.12, +0.42]**
Pure tones	−0.06 [−0.16, +0.03]	**+0.15 [+0.04, +0.27]**	−0.01 [−0.10, +0.08]	**+0.18 [+0.07, +0.30]**	**+0.19 [+0.06, +0.32]**
Pooled, harmonic (RE, $k = 6$)	−0.09 [−0.22, +0.03]	**+0.17 [+0.06, +0.29]**	−0.02 [−0.15, +0.11]	**+0.17 [+0.03, +0.31]**	**+0.18 [+0.04, +0.32]**

Table 6: *This study.* The out-of-sample screen (leave one US experiment out; point estimates). R^2: mean over held-out experiments of the fit to the whole profile (10 US; six harmonic). Gap, dip: left unexplained, mean of the six harmonic experiments. TT − other (TT, tritone): the dip left unexplained at the tritone minus its mean over the other 13 whole-number intervals from 1 to 14 semitones, all from the same held-out fit, averaged over the eight US experiments without stretched or compressed partials. Stretched: residual dip for stretched partials at 6.00 semitones and at the stretched scale's tritone, 6.42. SD units. Post-hoc rows: Supplementary Table S6.

	Held-out R^2					Stretched, dip left	
Model	10 US	6 harm.	Gap left	Dip left	TT − other	at 6.00	at 6.42
Existing models (weights refitted in each fold)							
Roughness (H&K)	0.476	0.569	+0.27	+0.25	+0.26	−0.04	+0.08
Harmonicity alone	0.135	0.248	+0.14	+0.07	+0.05	−0.03	+0.18
H&K + harmonicity ($\sigma = 6.83$ cents)	0.533	0.626	+0.20	+0.14	+0.14	−0.04	+0.08
Revised H&K + harmonicity	0.501	0.620	+0.25	+0.15	+0.19	−0.04	−0.01
Composite of Marjieh et al.	0.493	0.558	+0.18	+0.06	+0.08	−0.04	+0.05
Candidates listed before the screen was run							
H&K + harmonicity, tolerance σ chosen in fold	0.548	0.657	+0.19	+0.14	+0.15	−0.03	+0.09
H&K + harmonicity + root ambiguity	0.527	0.632	+0.21	+0.14	+0.15	−0.03	+0.08
H&K + harmonicity + periodicity	0.517	0.606	+0.20	+0.14	+0.14	−0.04	+0.08
H&K + periodicity	0.468	0.555	+0.26	+0.24	+0.25	−0.03	+0.08
H&K + harmonicity(σ) + ambiguity + periodicity	0.538	0.639	+0.17	+0.12	+0.13	−0.04	+0.09
H&K + harmonicity + harmonic entropy of f_0 ratio	0.519	0.630	+0.22	+0.14	+0.16	−0.08	+0.08

Table 7: *This study.* The four targeted tests, on the residual after roughness and harmonicity (published tolerance). SE: standard error, from the participant bootstrap (musicianship: random effects over eight experiments, Hartung–Knapp standard error, t distribution with 7 degrees of freedom). p: two-sided; Holm: adjusted over the four.

Contrast	Estimate	SE	p	Holm p
Culture: Korean − US, gap after rough. + harm. (harmonic tones)	−0.14	0.12	.227	.454
Training: musicians − non-musicians, gap after rough. + harm. (RE, $k = 8$)	+0.10	0.06	.121	.364
Third harmonic: observed − roughness-predicted change in the dip	+0.17	0.08	.045	.180
Roll-off: observed − roughness-predicted change in the gap, 2 vs 12 dB/oct	−0.04	0.11	.746	.746

Table 8: *This study.* Bowling et al. (2018): share (%) of the tritone–minor-sixth dyad difference reproduced by each fitted model (12 dyads), and the tritone penalty in chords (rating units per tritone, chord size and semitone count controlled [95% CI]; $n = 298$), with the models scored at equal-tempered (ET) or the presented just-intoned (JI) pitches.

	Dyad share (%)		Chord tritone penalty	
Model	ET	JI	ET	JI
Chord size only	–	–	+0.205	+0.205
Roughness (H&K)	−2	−2	+0.197 [+0.125, +0.268]	+0.198 [+0.125, +0.269]
Roughness (Sethares)	15	15	+0.184 [+0.115, +0.255]	+0.184 [+0.115, +0.255]
H&K + harmonicity	48	93	+0.080 [+0.006, +0.150]	+0.009 [−0.056, +0.074]
H&K + familiarity	44	44	+0.090 [+0.016, +0.161]	+0.094 [+0.020, +0.165]
H&K + harmonicity + familiarity	48	95	+0.002 [−0.070, +0.072]	−0.058 [−0.130, +0.009]
H&K + periodicity (Stolzenburg)	33	32	+0.109 [+0.037, +0.177]	+0.113 [+0.044, +0.178]